SAM3 / CHTC A100 Run Analysis

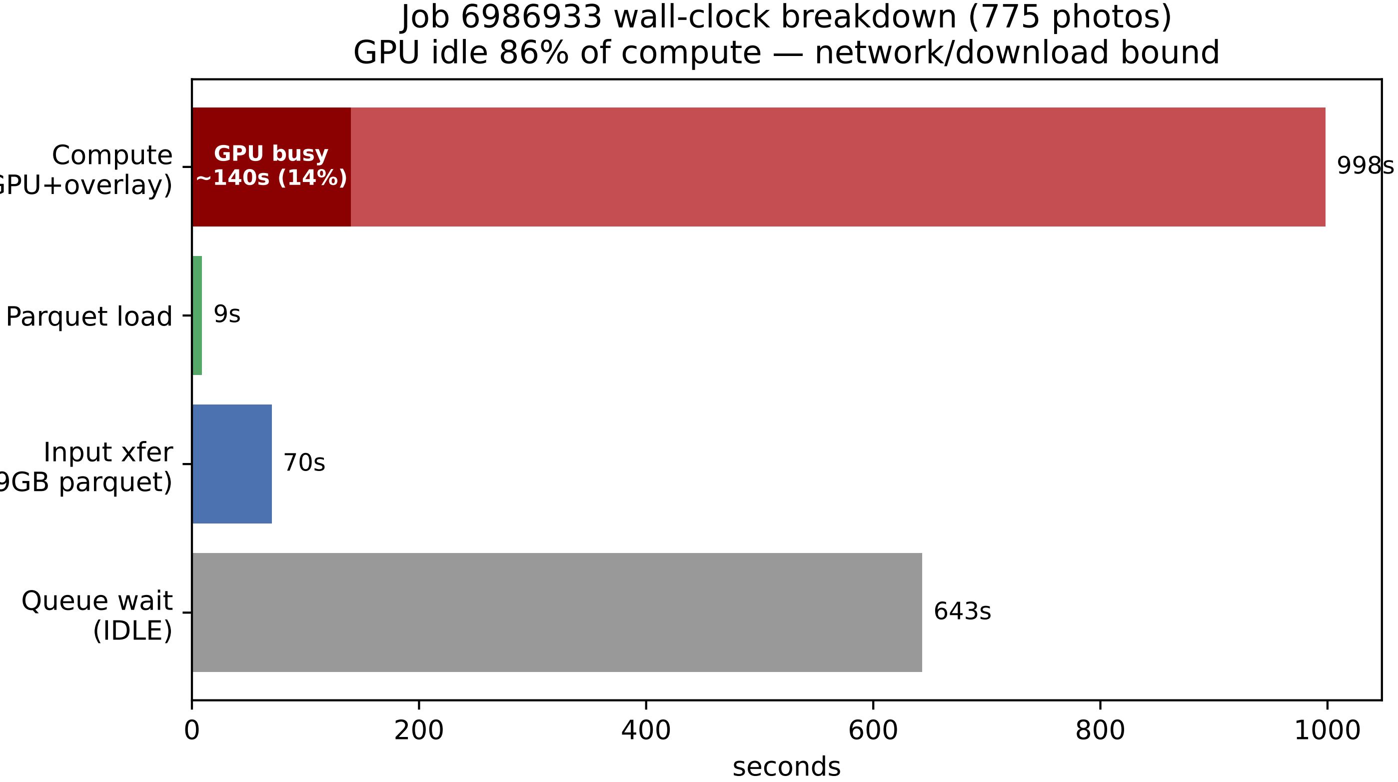
Job 6986933 | taxon 160559 | prompt='flower'

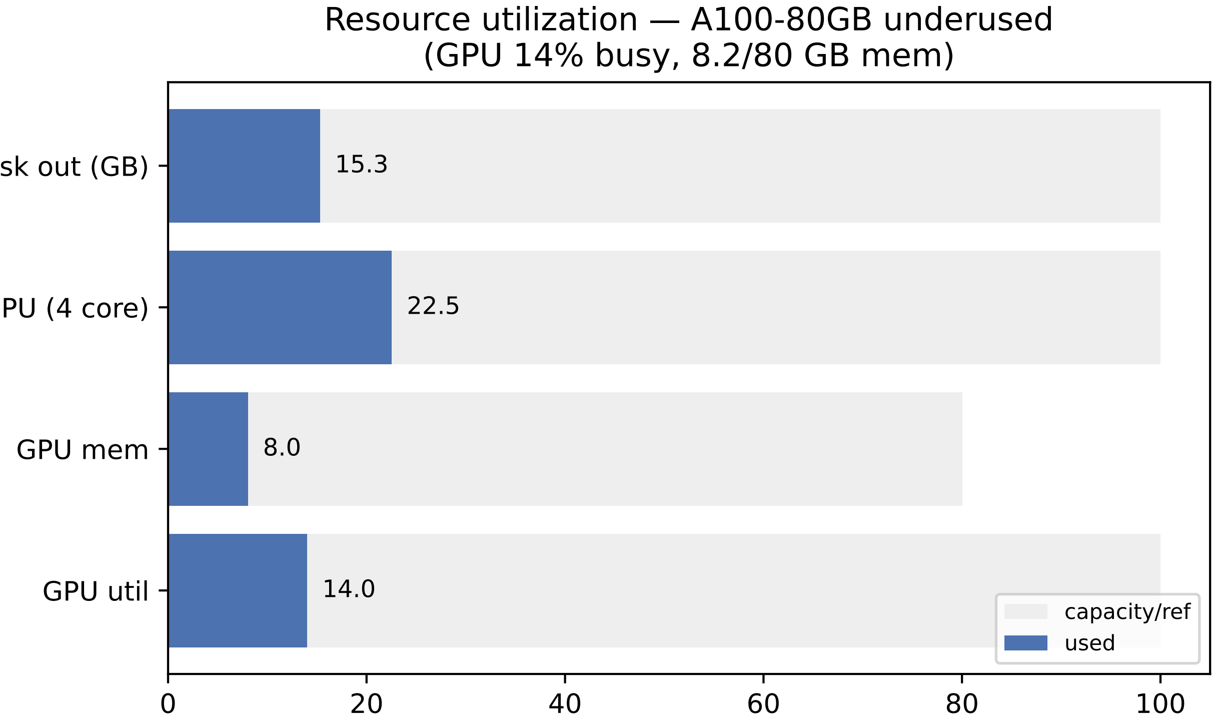
775 photos | 426 detected | 55.0% detection rate

generated by analyze.py

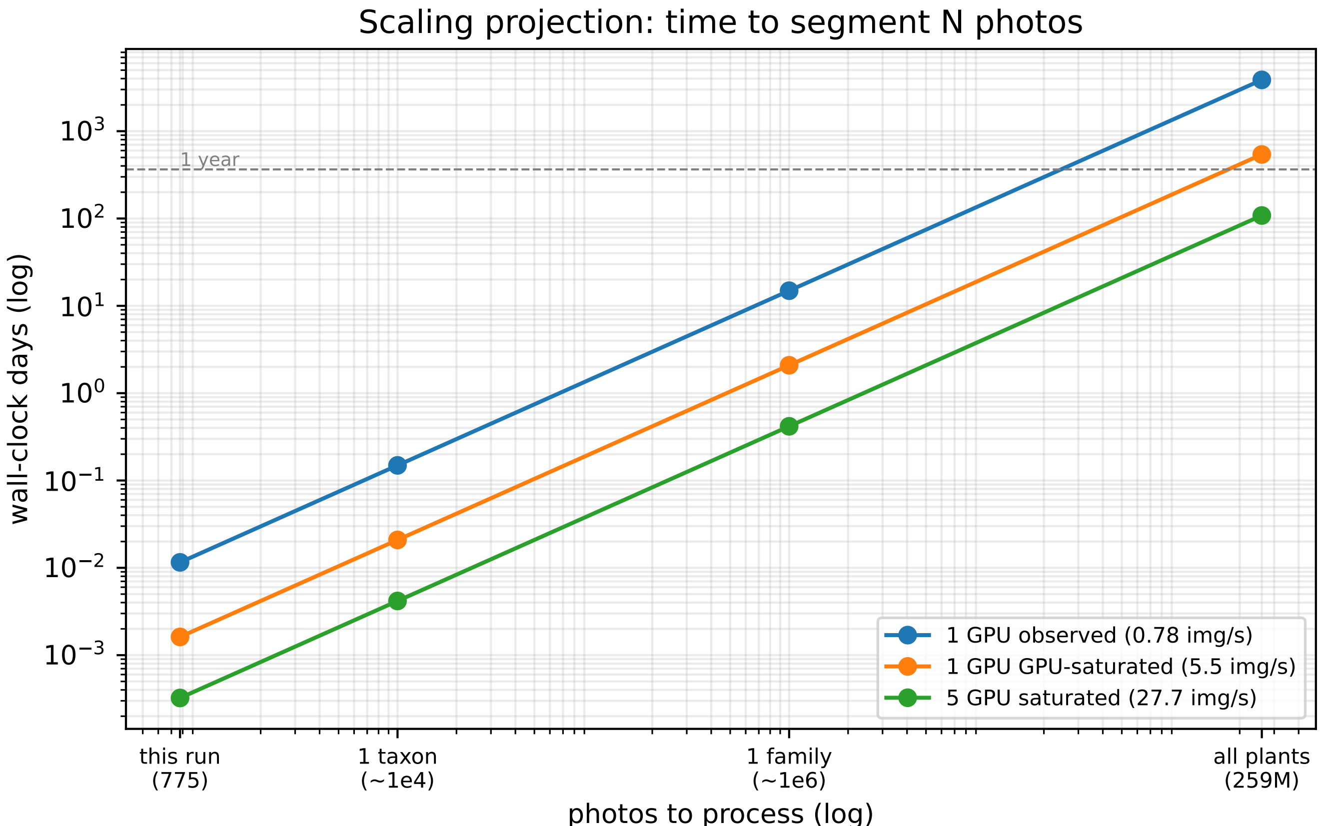
Summary metrics

```
photos                          : 775
detected_images                 : 426
no_detection                    : 349
detection_rate_%                : 55.0
total_masks                     : 1877
masks_per_detected_img          : 4.41
mean_conf_per_detection         : 0.9358
median_conf_per_detection       : 0.9343
mean_conf_per_image             : 0.942
compute_wall_s                  : 998
gpu_busy_s_est                  : 139.7
gpu_util_%                      : 14.000000000000002
throughput_observed_img_s       : 0.777
throughput_gpusat_img_s         : 5.55
output_MB_per_image             : 20.2
parquet_load_s                  : 8.786
observations                    : 269
images_per_obs_run              : 2.881
mean_conf_per_obs               : 0.9405
```

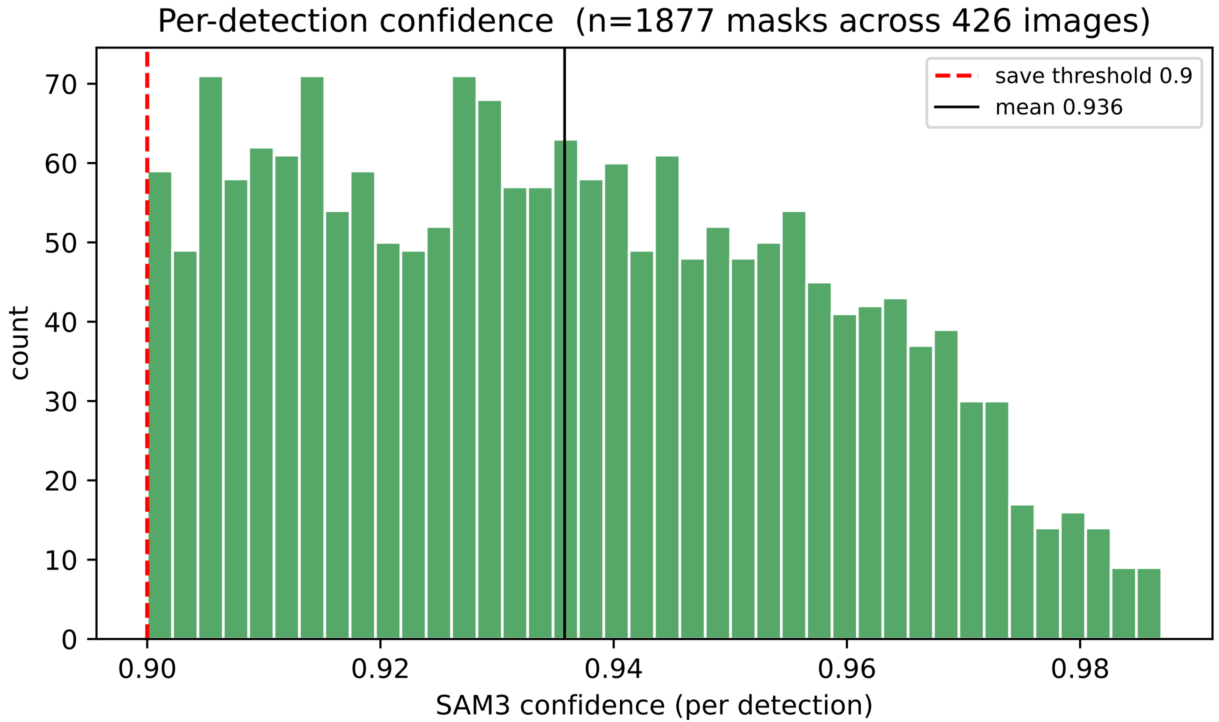

Job 6986933 wall-clock breakdown (775 photos)
GPU idle 86% of compute — network/download bound
Compute
GPU+overlay)
GPU busy
~140s (14%)
998s
Parquet load
9s
Input xfer
9GB parquet)
70s
Queue wait
(IDLE)
643s
0
200
400
600
800
1000
seconds

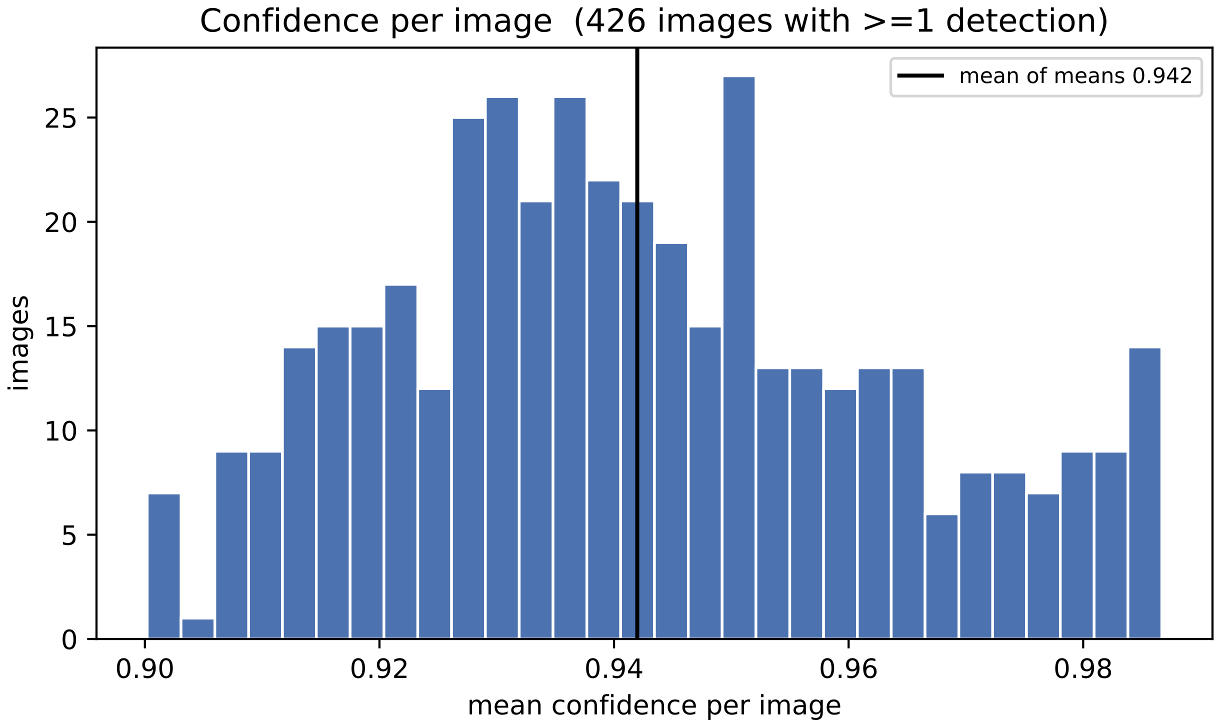


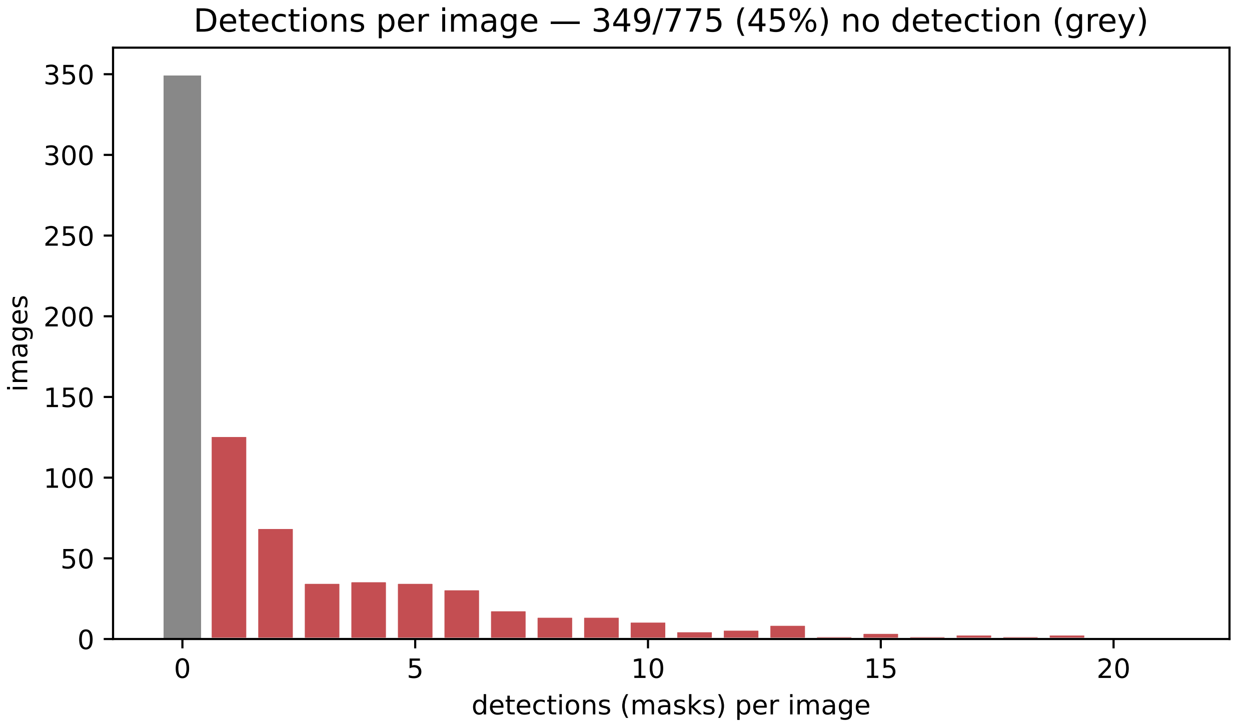
Resource utilization — A100-80GB underused
(GPU 14% busy, 8.2/80 GB mem)
sk out (GB)
PU (4 core)
GPU mem
GPU util
15.3
22.5
8.0
14.0
capacity/ref
used
0
20
40
60
80
100



Scaling projection: time to segment N photos
wall-clock days (log)
10^3
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
1 year
1 GPU observed (0.78 img/s)
1 GPU GPU-saturated (5.5 img/s)
5 GPU saturated (27.7 img/s)
this run (775)
1 taxon (~1e4)
1 family (~1e6)
all plants (259M)
photos to process (log)


Per-detection confidence (n=1877 masks across 426 images)
save threshold 0.9
mean 0.936
count
SAM3 confidence (per detection)
0
10
20
30
40
50
60
70
0.90
0.92
0.94
0.96
0.98


Confidence per image (426 images with >=1 detection)
mean of means 0.942
images
0
5
10
15
20
25
0.90
0.92
0.94
0.96
0.98
mean confidence per image


Detections per image — 349/775 (45%) no detection (grey)
images
0
50
100
150
200
250
300
350
0
5
10
15
20
detections (masks) per image

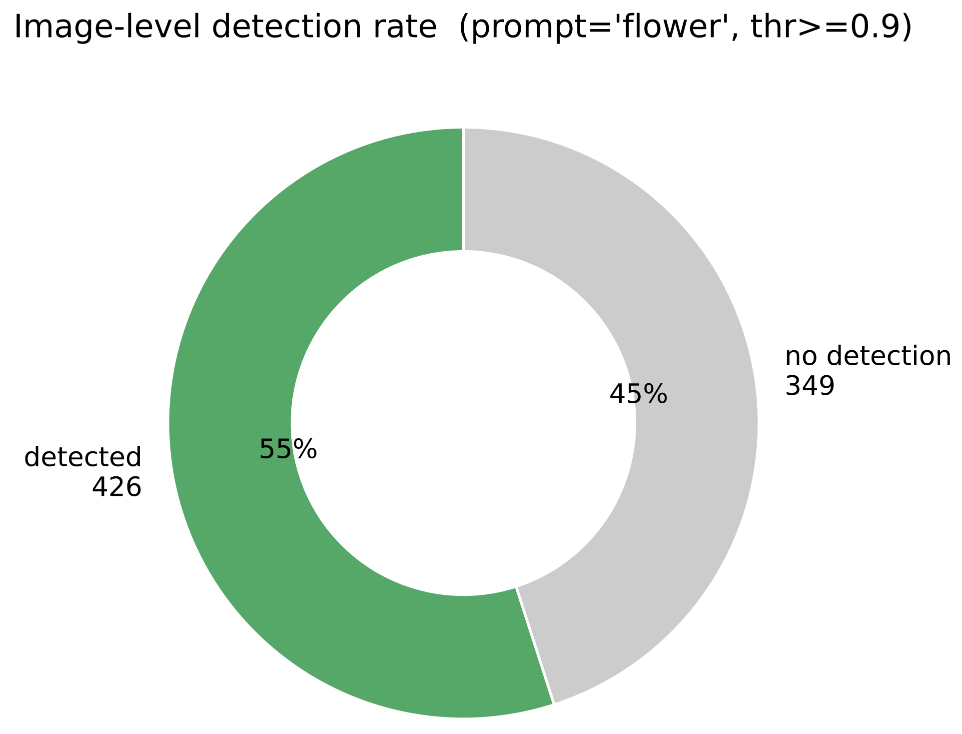
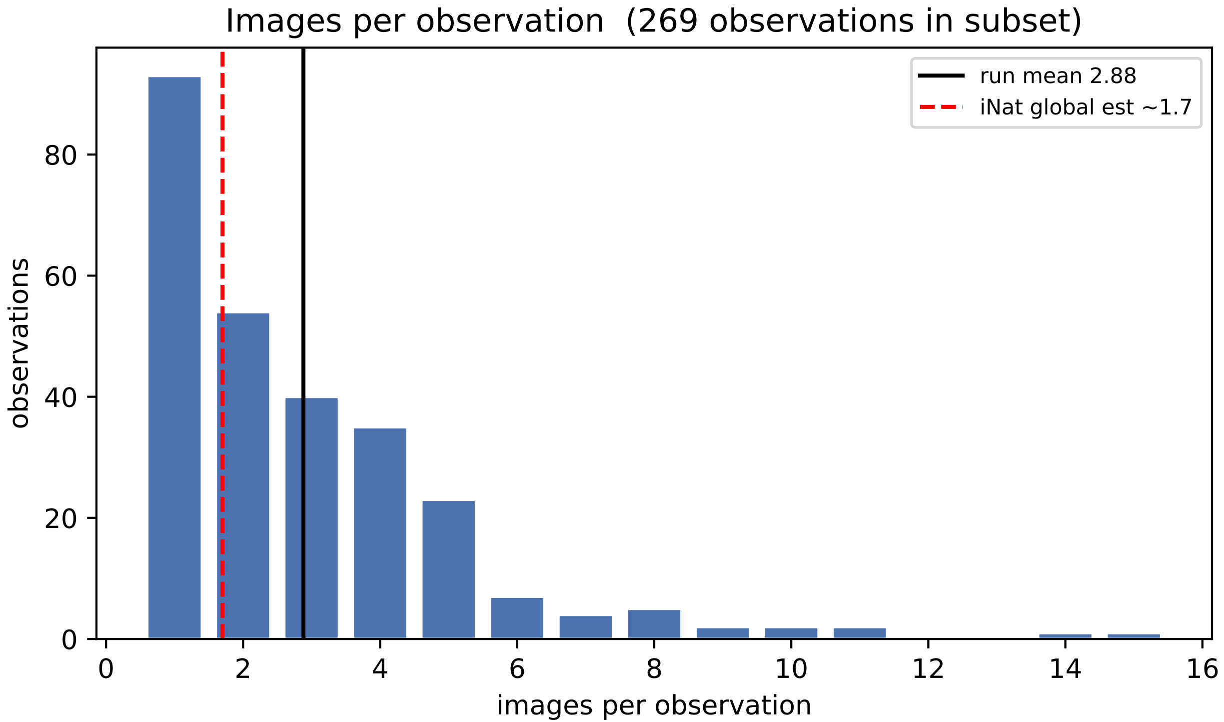
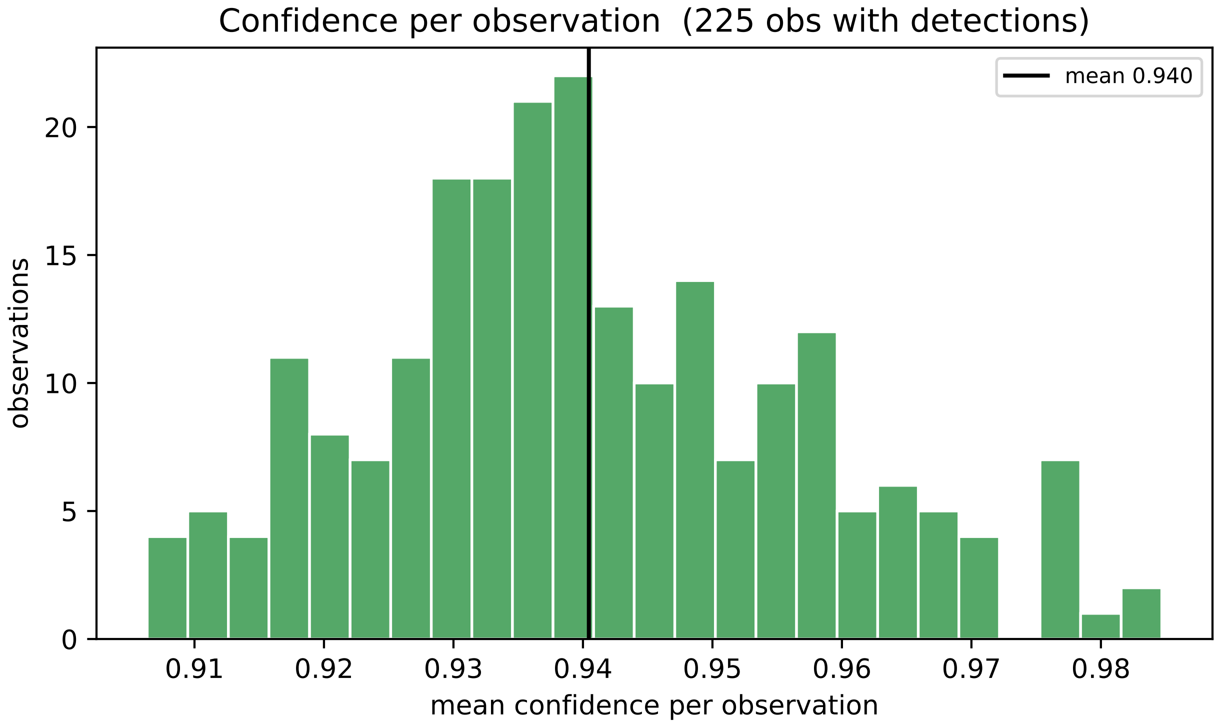
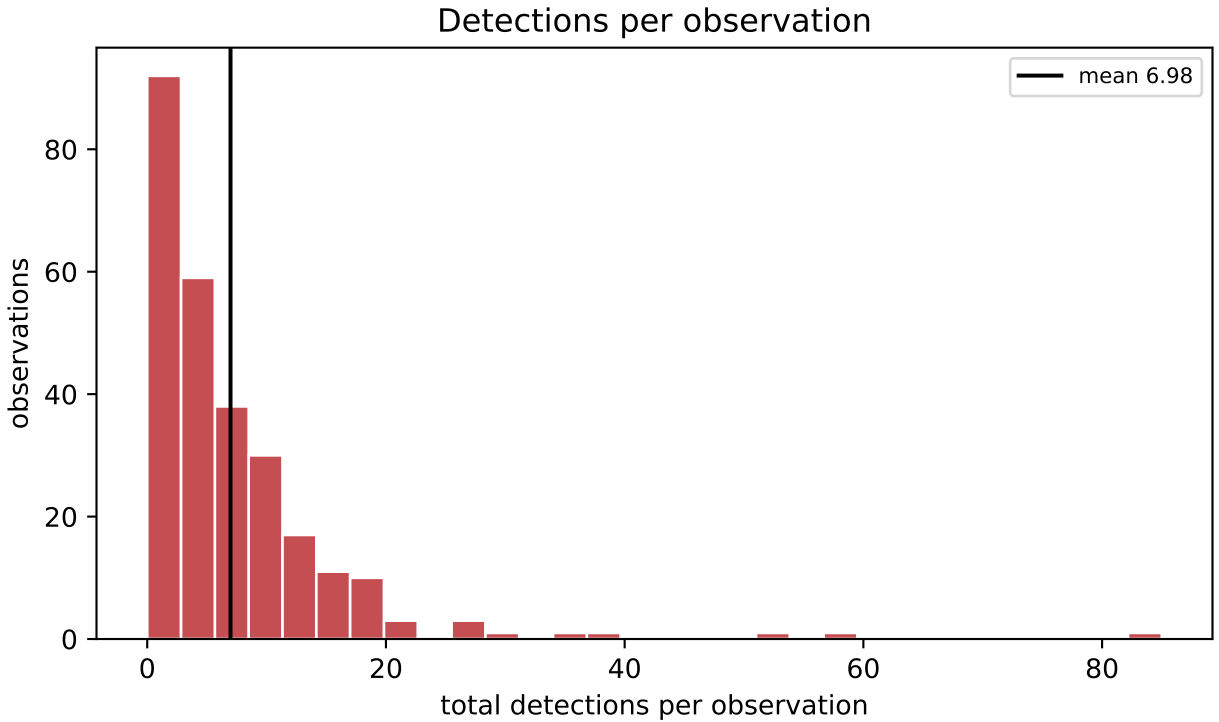
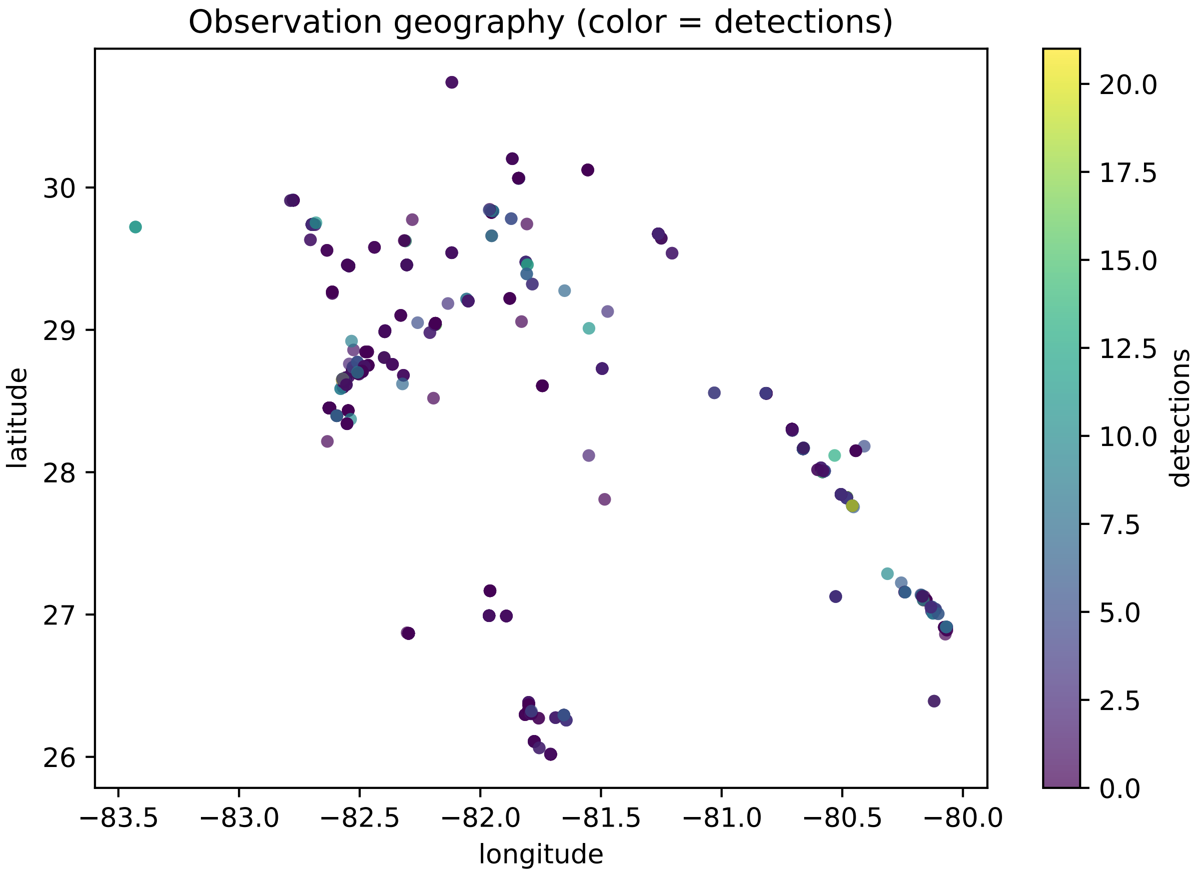


Image-level detection rate (prompt='flower', thr>=0.9)
no detection
349
45%
detected
426
55%


Images per observation (269 observations in subset)
run mean 2.88
iNat global est ~1.7
observations
images per observation
0
20
40
60
80
0
2
4
6
8
10
12
14
16


Confidence per observation (225 obs with detections)
mean 0.940
observations
0
5
10
15
20
0.91
0.92
0.93
0.94
0.95
0.96
0.97
0.98
mean confidence per observation


Detections per observation
mean 6.98
observations
total detections per observation
0
20
40
60
80
0
20
40
60
80



Observation geography (color = detections)
latitude
longitude
detections
30
29
28
27
26
−83.5
−83.0
−82.5
−82.0
−81.5
−81.0
−80.5
−80.0
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0

Scaling to all iNaturalist plants

```
Scaling target: 259,412,383 plant photos (inat_photos.parquet)

Throughput regimes (per the 775-photo run):
  observed end-to-end : 0.78 img/s/GPU  (network/download bound, GPU 14% busy)
  GPU-saturated       : 5.5 img/s/GPU  (if downloads kept GPU fed)

Wall-clock to process ALL plant photos:
  1 GPU  observed     : 3,866 d (10.6 yr)
  1 GPU  saturated    : 541 d (1.5 yr)
  5 GPU  saturated    : 108 d (0.3 yr)
  20 GPU saturated    : 27 d (0.1 yr)

Storage:
  output ~20.2 MB/image (masks .npy + segments + overlays + image)
  full scale ~ 4.9 PB  (!!)
  masks-only (.npy) dominate; 45% no-detect images add ~0 mask bytes

Bottleneck & scaling recommendations:
  1. NETWORK, not GPU. GPU idle 86%. Raise MAX_WORKERS (4 -> 16-32)
     download threads to feed the A100; ~5x headroom before GPU-bound.
  2. One A100 can serve many download workers; prefer more threads over
     more GPUs until util > ~70%.
  3. OUTPUT TRANSFER is the failure mode: 16 GB of many small .npy via
     Pelican timed out (err 6004) then 'already exists' on retry -> held.
     Fix: tar each shard's output to ONE archive before transfer.
  4. Store masks compressed (np.savez_compressed / packbits booleans):
     binary masks compress ~10-50x -> cuts the PB estimate hard.
  5. 45% no-detect at thr>=0.9 = wasted GPU+storage. Tune prompt/threshold,
     or pre-filter; consider lower thr with score kept in CSV for later cut.
  6. Shard by global_index % N across parallel A100 jobs (already designed).
```